INF10025 Data Management and Analytics

Task 2 – Credit Submission

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A. Task Credit2a

First and foremost, I had to download a file named ***Cust_Details.xlsx.*** In detail, the file contains two sheet named ***Sheet1*** and ***Sheet2.*** After that, I opened both sheets in Microsoft Access as two tables and changed table name to CUST5520 and CUST_TYPE5520 respectively.

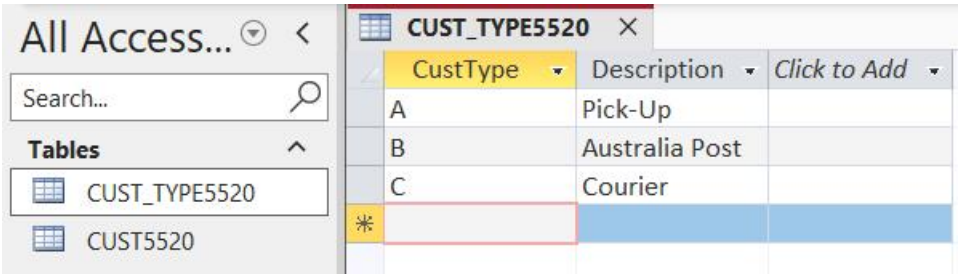


Figure 1: Data of CUST_TYPE5520 Table (Datasheet View).

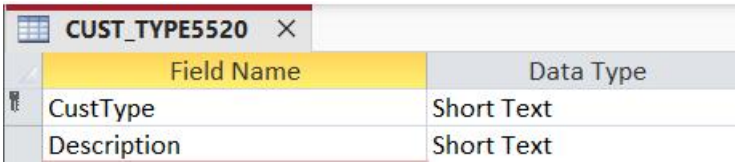


Figure 2: Design View of CUST_TYPE5520 Table.

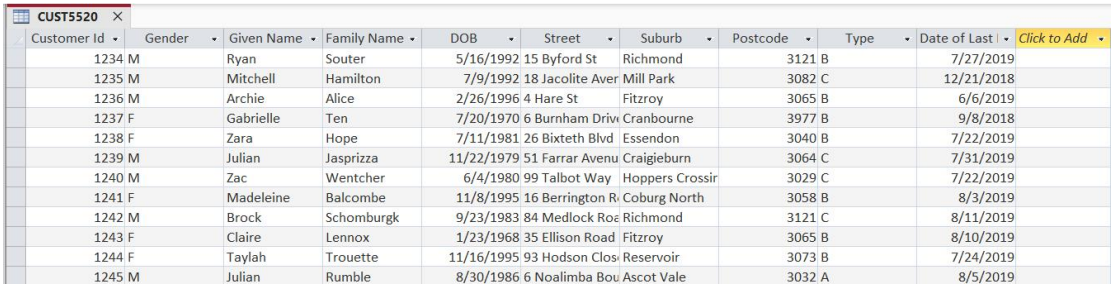


Figure 3: Data of CUST5520 Table (Datasheet View).

CUST5520

Field Name	Data Type
Customer Id	Number
Gender	Short Text
Given Name	Short Text
Family Name	Short Text
DOB	Date/Time
Street	Short Text
Suburb	Short Text
Postcode	Number
Type	Short Text
Date of Last Purchase	Date/Time

Figure 4: Design View of CUST5520 Table.

After renaming the two tables above, I established relationships between them based on Primary Key and Foreign Key. Based on the Design View of the two tables, I decided to link ***CustType (Primary Key)*** field in the ***CUST_TYPE5520*** table with ***Type (Foreign Key)*** field in the ***CUST5520*** table.

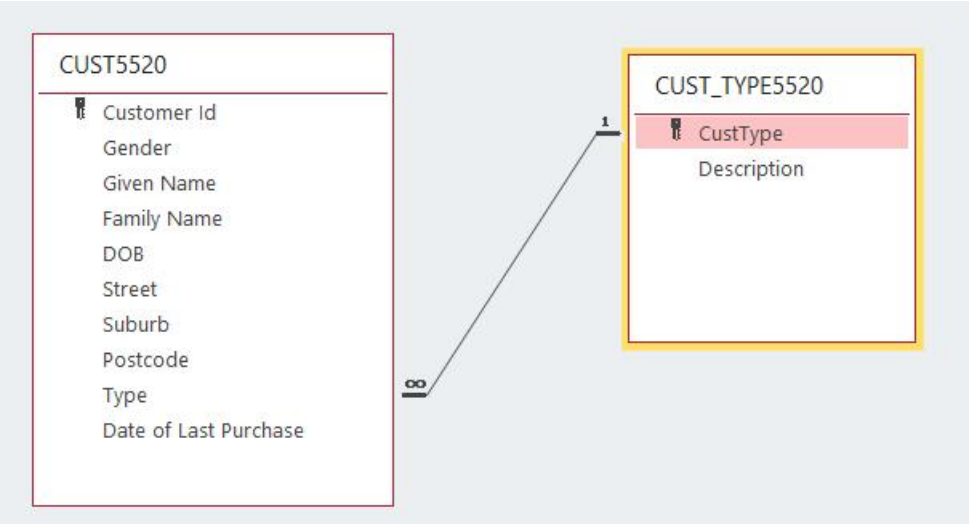


Figure 5: Relationship between CUST5520 and CUST_TYPE5520 Tables.

It is clear that one ***CustType*** can have many records/rows of ***Type*** in the ***CUST5520*** table, and one record in the ***CUST5520*** table corresponds to only one specific ***CustType*** in the ***CUST_TYPE5520*** table, establishing a one-to-many relationship.

B. Task Credit2b

The sfunction yntax of the DateDiff in Microsoft Access is:

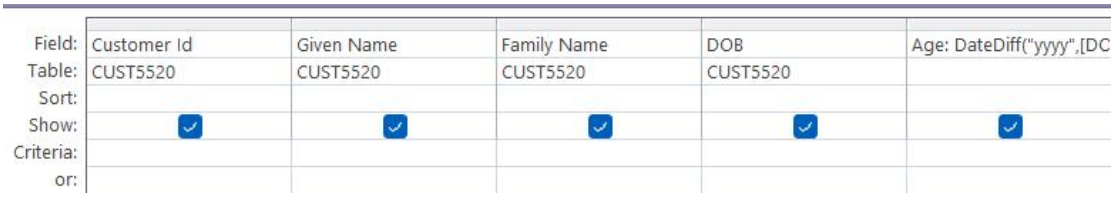
DateDiff(interval, date1, date2[, firstdayofweek][, firstweekofyear])

- **interval (mandatory):** the time unit to use for calculating the distance (enclosed in quotation marks), including 'yyyy', 'q', 'm', and so on.
- **date1 and date2 (mandatory):** two timestamps to compare. If 'date1' is greater than 'date2', the function returns a negative value.
- **firstdayofweek (Optional):** defines the first day of the week (Default: Sunday, which means 1).
- **firstweekofyear (Optional):** defines the first week of the year (Default: the week containing January 1st, which means 1).

In terms of calculating the age of customer, the method is to subtract the customer's birth year from current year. Subsequently, the formula for the ***Age*** column is:

Age: DateDiff("yyyy",[DOB],Now())

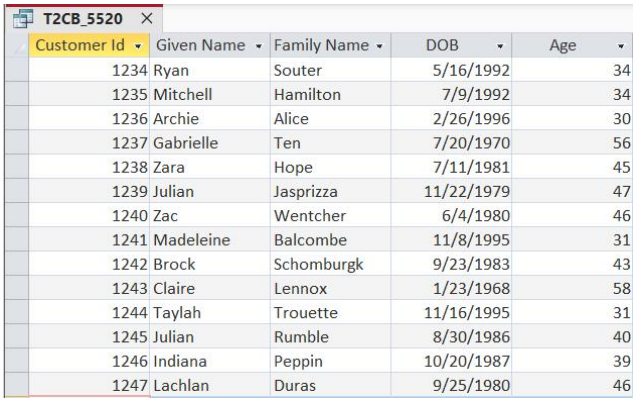
In the above formula, I created a new column named ***Age*** first. In detail, the ***"yyyy"*** parameter requires Access to calculate the difference based on years, ***[DOB]*** is the date of birth field, and ***Now()*** retrieves the current system date.



Field:	Customer Id	Given Name	Family Name	DOB	Age: DateDiff("yyyy",[DC
Table:	CUST5520	CUST5520	CUST5520	CUST5520	
Sort:					
Show:	☑	☑	☑	☑	☑
Criteria:					
or:					

Figure 6: Design View of T2CB_5520 Query.

After completing the design grid of the query named T2CB_5520, I clicked the ***Run*** button to view the results.



T2CB_5520

Customer Id	Given Name	Family Name	DOB	Age
1234	Ryan	Souter	5/16/1992	34
1235	Mitchell	Hamilton	7/9/1992	34
1236	Archie	Alice	2/26/1996	30
1237	Gabrielle	Ten	7/20/1970	56
1238	Zara	Hope	7/11/1981	45
1239	Julian	Jasprizza	11/22/1979	47
1240	Zac	Wentcher	6/4/1980	46
1241	Madeleine	Balcombe	11/8/1995	31
1242	Brock	Schomburgk	9/23/1983	43
1243	Claire	Lennox	1/23/1968	58
1244	Taylah	Trouette	11/16/1995	31
1245	Julian	Rumble	8/30/1986	40
1246	Indiana	Peppin	10/20/1987	39
1247	Lachlan	Duras	9/25/1980	46

Figure 7: Datasheet View of T2CB_5520 Query Results.

Settings	*Description*
yyyy	Year
q	Quarter
m	Month
y	Day of the year
d	Day
w	Weekday
ww	Week
h	Hour
n	Minute
s	Second

Figure 8: DateDiff Interval Settings.

Constant	*Value*	*Description*
vbUseSystem	0	Use NLS API settings.
vbSunday	1	Sunday
vbMonday	2	Monday
vbTuesday	3	Tuesday
vbWednesday	4	Wednesday
vbThursday	5	Thursday
vbFriday	6	Friday
vbSaturday	7	Saturday

Figure 9: DateDiff firstdayofweek Settings.

C. Task Credit2c

In the Task Credit2c description, this section requires students to generate a new query named ***T2CC_5520*** using the previous query of task ***Credit2b*** as it sources.Next, I dragged and dropped all columns from the ***T2CB_5520*** query and filled in the Criteria row of the Age column with: '***<= [Enter Age]***'



Figure 10: Design View of T2CC_5520 Parameter Query

After I clicked the ***Run*** button, a dialog box appeared, prompting me to input the age value.

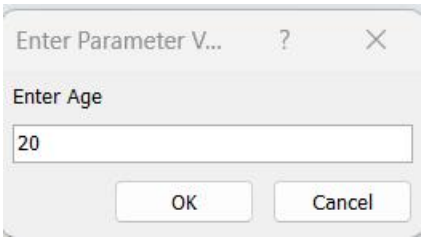


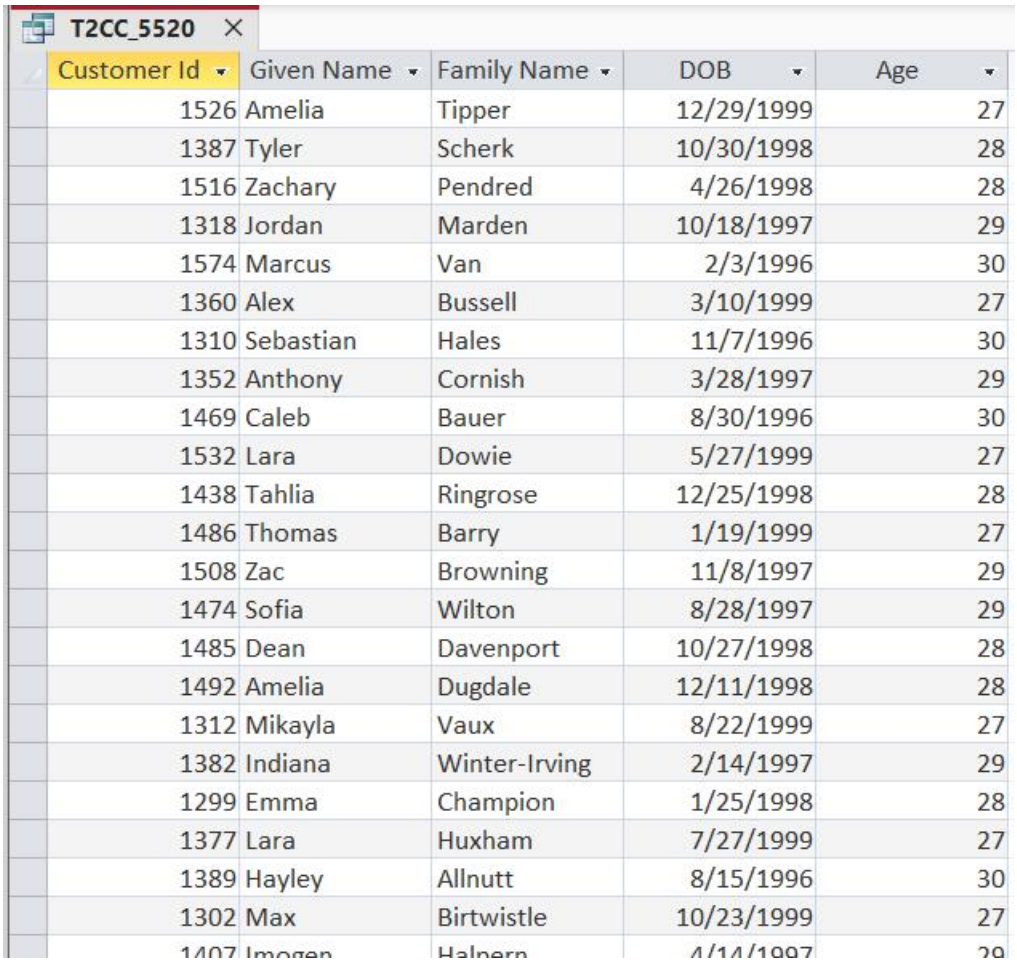
Figure 11: Input Dialog Box Prompting for Age Value

As a result, the query searched customers whose age is equal to or less than 20 and the results were displayed in the ***Datasheet View***.



Figure 12: Datasheet View of T2CC_5520 Parameter Query Results

Based on Figure 12, it is clear that no records matched the condition of being 20 years old or less. Now, I will try filtering with a limit of 30 years old.



T2CC_5520

Customer Id	Given Name	Family Name	DOB	Age
1526	Amelia	Tipper	12/29/1999	27
1387	Tyler	Scherk	10/30/1998	28
1516	Zachary	Pendred	4/26/1998	28
1318	Jordan	Marden	10/18/1997	29
1574	Marcus	Van	2/3/1996	30
1360	Alex	Bussell	3/10/1999	27
1310	Sebastian	Hales	11/7/1996	30
1352	Anthony	Cornish	3/28/1997	29
1469	Caleb	Bauer	8/30/1996	30
1532	Lara	Dowie	5/27/1999	27
1438	Tahlia	Ringrose	12/25/1998	28
1486	Thomas	Barry	1/19/1999	27
1508	Zac	Browning	11/8/1997	29
1474	Sofia	Wilton	8/28/1997	29
1485	Dean	Davenport	10/27/1998	28
1492	Amelia	Dugdale	12/11/1998	28
1312	Mikayla	Vaux	8/22/1999	27
1382	Indiana	Winter-Irving	2/14/1997	29
1299	Emma	Champion	1/25/1998	28
1377	Lara	Huxham	7/27/1999	27
1389	Hayley	Allnutt	8/15/1996	30
1302	Max	Birtwistle	10/23/1999	27
1407	Imogen	Halpern	4/14/1997	29

Figure 13: Datasheet View of T2CC_5520 Results with Age Limit Set to 30

After viewing data, I could say that the minimum age of custoner in the database is 27 years old.

D. Task Credit2d

In the fourth task, I had to use both tables ***CUST5520*** and ***CUST_TYPE5520*** to create a query named ***T2CD_5520***. Next, I selected the ***Totals*** function to show the ***Total*** row and dragged and dropped the following columns into the ***Design Grid:***

- ***Field***: Suburb, ***Table***: CUST5520, ***Total***: Group By, ***Sort***: Ascending
- ***Field***: CustType, ***Table***: CUST_TYPE5520, ***Total***: Group By, ***Sort***: Ascending
- ***Field***: Description, ***Table***: CUST_TYPE5520, ***Total***: Group By, ***Sort***: Ascending
- ***Field***: Customer Id, ***Table***: CUST5520, ***Total***: Count, ***Sort***: Descending

Field:	Suburb	CustType	Description	Customer Id
Table:	CUST5520	CUST_TYPE5520	CUST_TYPE5520	CUST5520
Total:	Group By	Group By	Group By	Count
Sort:	Ascending	Ascending	Ascending	Descending
Show:	☑	☑	☑	☑
Criteria:				
or:				

Figure 14: Design View of T2CD_5520 Query

T2CD_5520

Suburb	CustType	Description	CountOfCust
Ascot Vale	A	Pick-Up	5
Ascot Vale	B	Australia Post	2
Ascot Vale	C	Courier	2
Brooklyn	B	Australia Post	4
Brooklyn	C	Courier	3
Brunswick	A	Pick-Up	3
Brunswick	B	Australia Post	5
Brunswick	C	Courier	5
Bundoora	A	Pick-Up	1
Bundoora	B	Australia Post	2
Bundoora	C	Courier	1
Carlton	A	Pick-Up	2
Carlton	B	Australia Post	2
Carlton	C	Courier	4
Chadstone	A	Pick-Up	2
Chadstone	B	Australia Post	2
Chadstone	C	Courier	2
Coburg North	A	Pick-Up	3

Figure 15: Datasheet View of T2CD_5520 Query

E. Task Credit2e

In terms of exporting to a text file named T2CD_5520.txt, I chose ***Text File*** in the ***External Data*** tab, selected the storage location, clicked ***Delimited*** (Characters such as comma or tab separate each field), ticked ***Include Field Names on First Row***, and chose ***Comma*** as the delimiter with double quotes as the text qualifier.

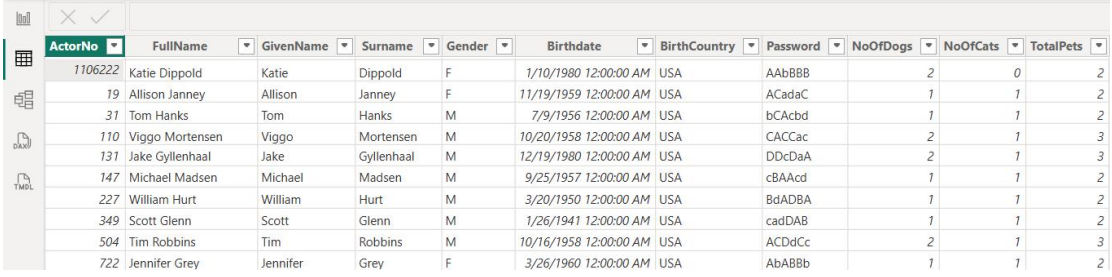
```
"Suburb","CustType","Description","CountOfCustomer Id"
"Ascot Vale","A","Pick-Up",5
"Ascot Vale","B","Australia Post",2
"Ascot Vale","C","Courier",2
"Brooklyn","B","Australia Post",4
"Brooklyn","C","Courier",3
"Brunswick","A","Pick-Up",3
"Brunswick","B","Australia Post",5
"Brunswick","C","Courier",5
"Bundoora","A","Pick-Up",1
"Bundoora","B","Australia Post",2
"Bundoora","C","Courier",1
"Carlton","A","Pick-Up",2
"Carlton","B","Australia Post",2
"Carlton","C","Courier",4
"Chadstone","A","Pick-Up",2
"Chadstone","B","Australia Post",2
"Chadstone","C","Courier",2
"Coburg North","A","Pick-Up",3
"Coburg North","B","Australia Post",3
"Craigieburn","B","Australia Post",3
"Craigieburn","C","Courier",1
"Cranbourne","A","Pick-Up",3
```

Figure 16: First 10 Rows of the Exported T2CD_5520.txt File

F. Task Credit2f

In the description of the task Credit2f, first and foremost, I selected the ***ACTOR5520*** table in ***Data View*** mode. Then, I added a new column named ***TotalPets*** using the formula:

TotalPets = ACTOR5520[NoOfDogs] + ACTOR5520[NoOfCats]



ActorNo	FullName	GivenName	Surname	Gender	Birthdate	BirthCountry	Password	NoOfDogs	NoOfCats	TotalPets
1106222	Katie Dippold	Katie	Dippold	F	1/10/1980 12:00:00 AM	USA	AAbBBB	2	0	2
19	Allison Janney	Allison	Janney	F	11/19/1959 12:00:00 AM	USA	ACadaC	1	1	2
31	Tom Hanks	Tom	Hanks	M	7/9/1956 12:00:00 AM	USA	bCAcbd	1	1	2
110	Viggo Mortensen	Viggo	Mortensen	M	10/20/1958 12:00:00 AM	USA	CACCac	2	1	3
131	Jake Gyllenhaal	Jake	Gyllenhaal	M	12/19/1980 12:00:00 AM	USA	DDcDaA	2	1	3
147	Michael Madsen	Michael	Madsen	M	9/25/1957 12:00:00 AM	USA	cBAAcd	1	1	2
227	William Hurt	William	Hurt	M	3/20/1950 12:00:00 AM	USA	BdADBA	1	1	2
349	Scott Glenn	Scott	Glenn	M	1/26/1941 12:00:00 AM	USA	cadDAB	1	1	2
504	Tim Robbins	Tim	Robbins	M	10/16/1958 12:00:00 AM	USA	ACDdCc	2	1	3
722	Jennifer Grey	Jennifer	Grey	F	3/26/1960 12:00:00 AM	USA	AbABBb	1	1	2

Figure 17: ***Data View of ACTOR5520 Table Showing the New TotalPets Column***

Regarding the visualization, I decided to create two charts: a **Clustered Column Chart** named *Total Pets by Gender* and a **Map** named *Average Pets per Country*.

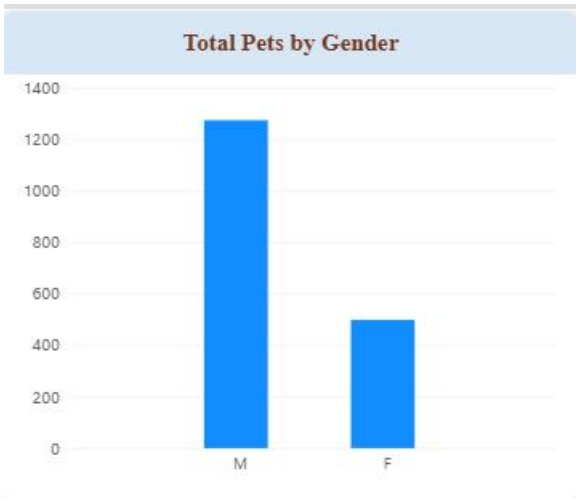


Figure 18: Clustered Column Chart Showing Total Pets by Gender



Figure 19: Map Visualization of Average Pets per Country

It can be seen from the Clustered Column Chart (Figure 18) that males have more pets, including cats and dogs, compared to females. Moreover, the distribution of pets on the map showcases that North America, particularly the United States, holds the highest concentration of actors and pet ownership while other regions across Europe, Asia, and Australia display smaller, dispersed bubbles indicating lower average pet numbers per country.

G. Task Credit2g

Subsequently, this task requires students to create a new column named DAYNO_5520 for calculating the day of the week based on each actor's date of

birth. Then, I selected the ACTOR5520 table in Data View, added a new column, and applied the following syntax:

DAYNO_5520 = WEEKDAY(ACTOR5520[Birthdate], 1)



```
1 DayNo_5520 = WEEKDAY(ACTOR5520[Birthdate], 1)
```

ActorNo	FullName	GivenName	Surname	Gender	Birthdate	BirthCountry	Password	NoOfDogs	NoOfCats	TotalPets	DayNo_5520
10	Bob Peterson	Bob	Peterson	M	1/18/1961 12:00:00 AM	USA	dDDbAb	0	0	0	4
139	Uma Thurman	Uma	Thurman	F	4/29/1970 12:00:00 AM	USA	DBCAdA	0	0	0	4
539	Thomas Lennon	Thomas	Lennon	M	8/9/1970 12:00:00 AM	USA	Dacadb	0	0	0	1
882	Liv Tyler	Liv	Tyler	F	7/1/1977 12:00:00 AM	USA	BAbBba	0	0	0	6
1064	Crispin Glover	Crispin	Glover	M	4/20/1964 12:00:00 AM	USA	dDbaba	0	0	0	2
1892	Matt Damon	Matt	Damon	M	10/8/1970 12:00:00 AM	USA	cBacCD	0	0	0	5
1895	Carl Reiner	Carl	Reiner	M	3/20/1922 12:00:00 AM	USA	bDAaCC	0	0	0	2
2505	James Cromwell	James	Cromwell	M	1/27/1940 12:00:00 AM	USA	CdDDDc	0	0	0	7
2975	Laurence Fishburne	Laurence	Fishburne	M	7/30/1961 12:00:00 AM	USA	DBbBBc	0	0	0	1
3141	Marisa Tomei	Marisa	Tomei	F	12/4/1964 12:00:00 AM	USA	CCCbBD	0	0	0	6
3197	Tom Sizemore	Tom	Sizemore	M	11/29/1961 12:00:00 AM	USA	cAAbaA	0	0	0	4
3636	Paul Newman	Paul	Newman	M	1/26/1925 12:00:00 AM	USA	ddbAAA	0	0	0	2
4090	Shirley MacLaine	Shirley	MacLaine	F	4/24/1934 12:00:00 AM	USA	AbAAcc	0	0	0	3
4139	Tom Skerritt	Tom	Skerritt	M	8/25/1933 12:00:00 AM	USA	BCcAdc	0	0	0	6
4299	Dean Martin	Dean	Martin	M	6/7/1917 12:00:00 AM	USA	bacACa	0	0	0	5
5292	Denzel Washington	Denzel	Washington	M	12/28/1954 12:00:00 AM	USA	bAabBB	0	0	0	3

Figure 20: Data View of ACTOR5520 Table Showing the New DayNo_5520 Column

Furthermore, the task requires me to generate a donut chart. Specifically, I placed the ***DayNo_5520*** field into the ***Legend*** and the ***ActorNo*** field into the ***Values*** to analyze the distribution. In additiion, I set title for the donurt chart is *Total Actors by Birth Day Number* with ***Font Size*** is ***15, Color*** is ***Brown***, ***Bold*** and ***Center Alignment.***

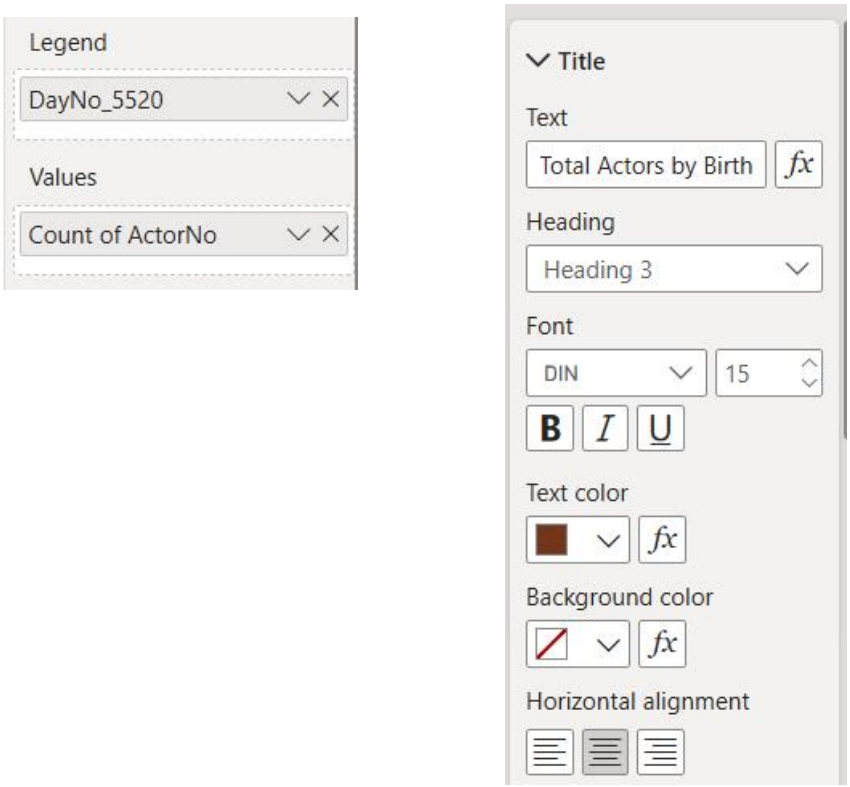


Figure 21: Build Visual and General Configuration

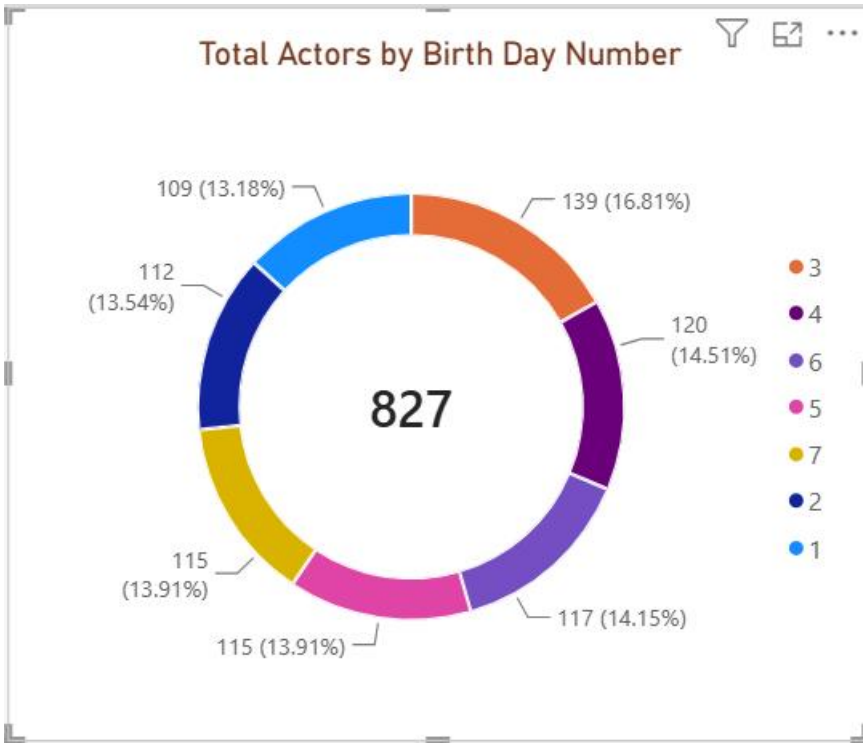


Figure 22: Total Actors by Birth Day Number Chart

H. Task Credit2h

First and foremost, I switched to ***Data View*** mode and clicked Enter Data on the ***Home*** tab, which opened the ***Create Table dialog box***. Then, I entered ***DayNo_5520*** for ***Column1*** and ***DayName*** for ***Column2***. Next, I populated the rows with values ranging from 1 (Sunday) to 7 (Saturday) respectively. After that, I clicked on Load button and renamed table to WeekDays5520

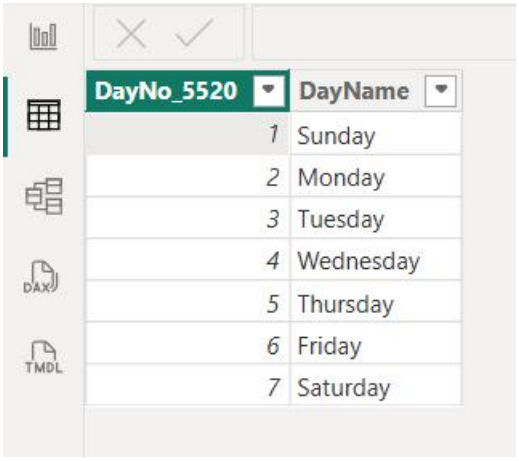


Figure 23: Data View of the WeekDays5520 Table

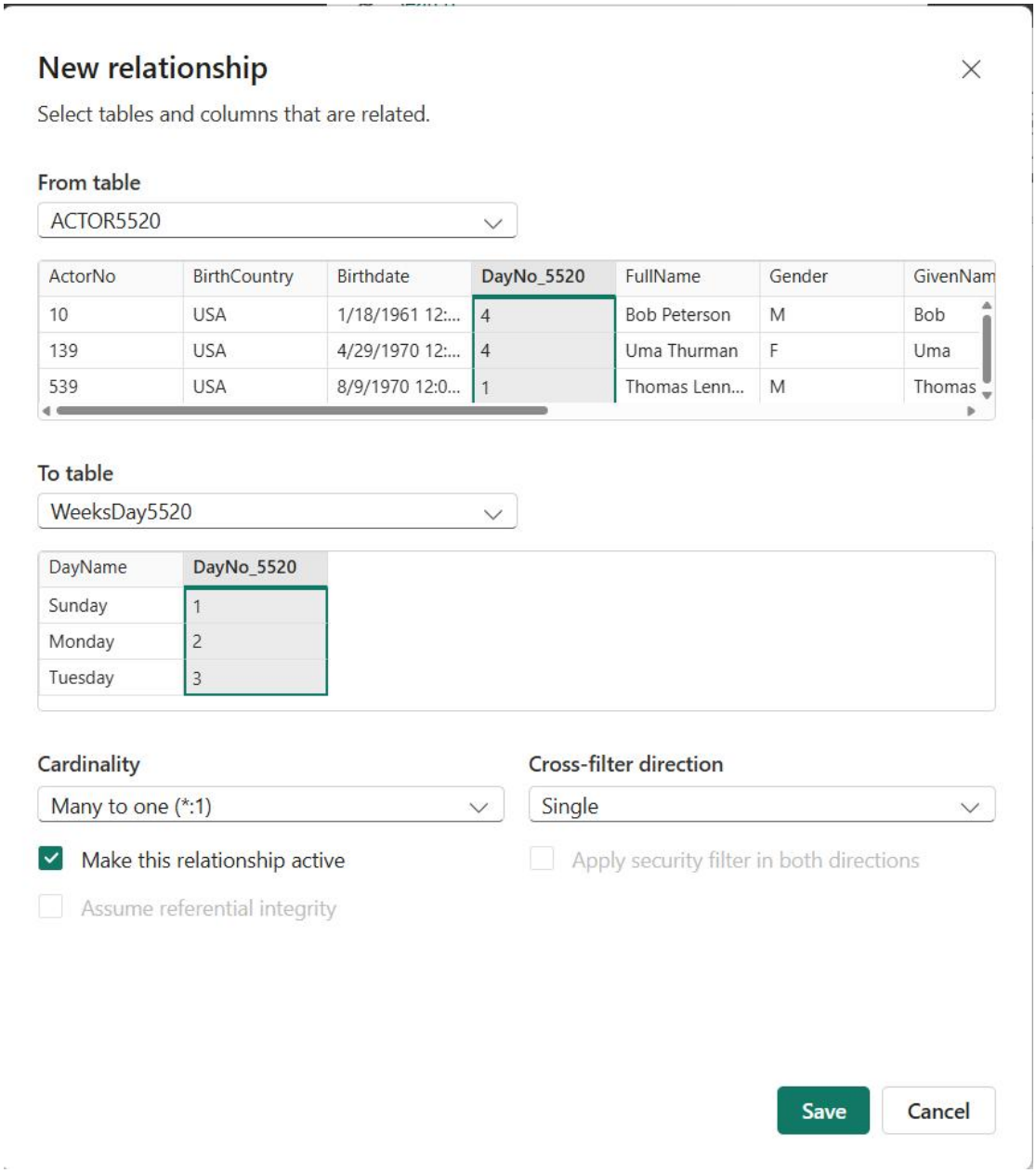


Figure 24: New Relationship Dialog Box between ACTOR5520 and WeeksDay5520

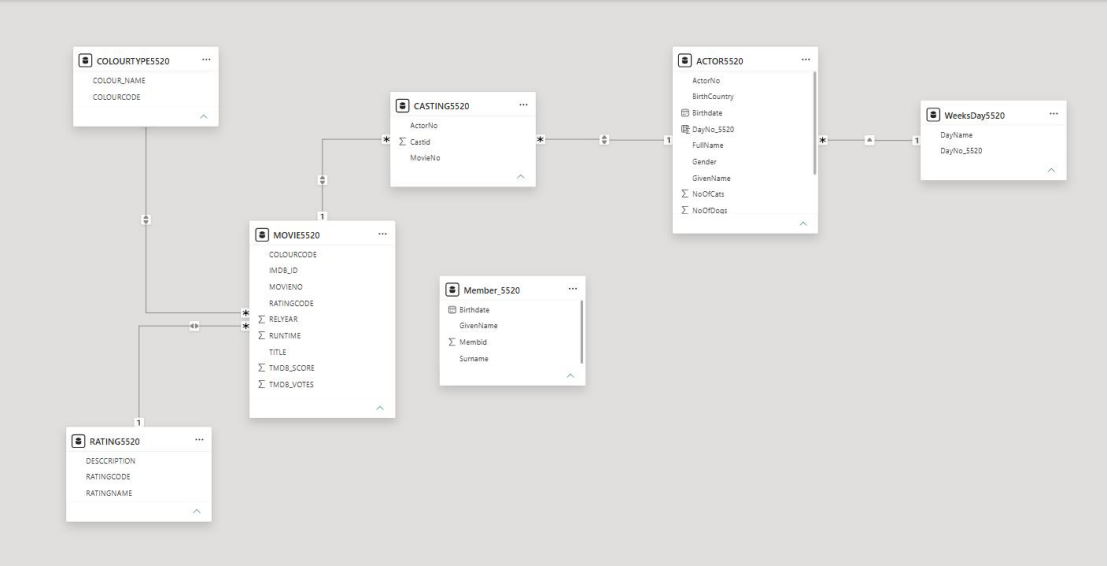


Figure 25: Relationships View of the Database Model

I. Task Credit2i

In Task 2i, I was required to create a pie chart to showcase the relationship between actors and their birth day names, illustrating the distribution of the ***Count of actors*** by ***Birth DayName***. Additionally, I placed the ***DayName*** field into the ***Legend*** and Count of ActorNo into the **Values**, set the chart title to ***Total Actors by Birth Day Name***, and applied styling consistent with the previous charts.

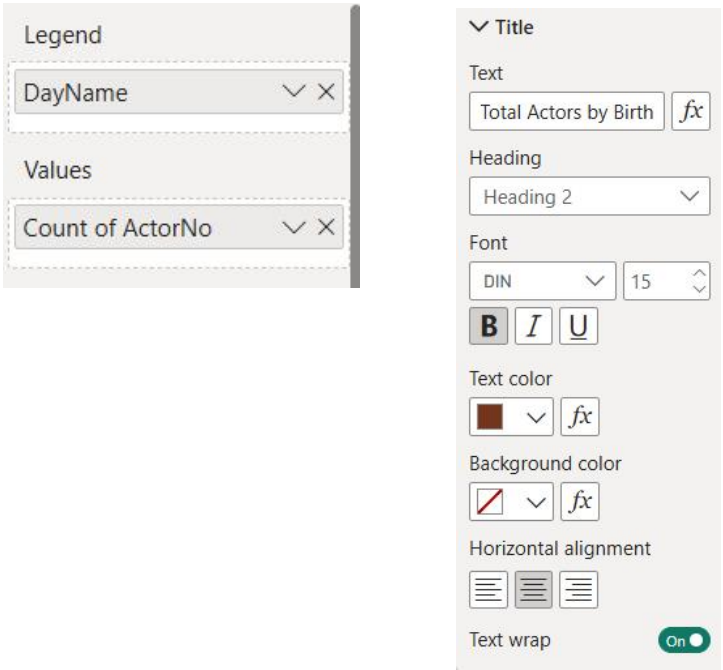


Figure 26: Build Visual and Format Visual Configuration for the Pie Chart

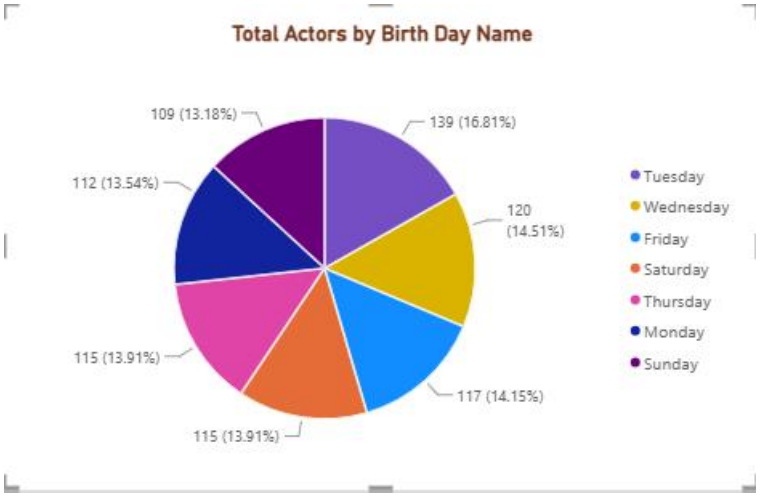


Figure 27: Pie Chart Showing Total Actors by Birth Day Name

J. Task Credit2j

In the final task of the Credit section, I added a Slicer visualization using the ***DayName*** field from the ***WeeksDay5520*** table and titled it ***Filter by Day Name***. Therefore, this interactive slicer allows users to filter the entire dashboard by specific days of the week, seamlessly integrating all four visual elements for dynamic data analysis.



Figure 28: Slicer for filtering data by Day Name

Finally, I added a new Text Box containing my full name (Truong Ngoc Gia Hieu), student ID (105565520), and a brief description of the report's purpose and interactive features which ensures all individual identification and documentation requirements are fully met on the dashboard.

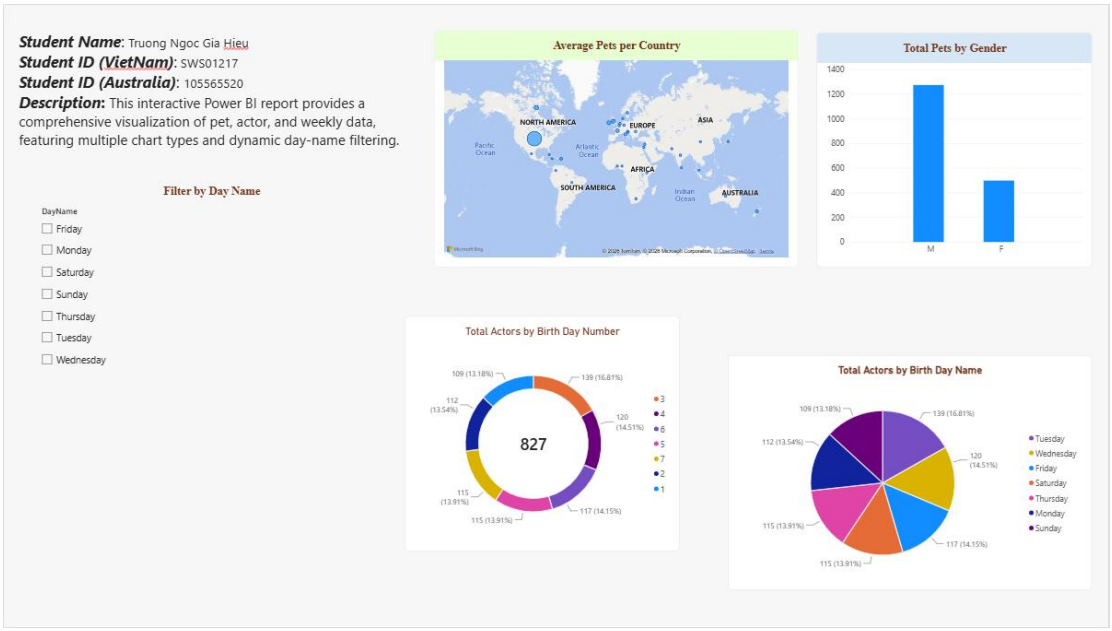


Figure 29: Final Power BI Report Dashboard

K. Conclusion

In this report, a comprehensive and interactive Power BI dashboard was successfully developed. By integrating multiple visual types (bar chart, map, donut chart, and pie chart), a dynamic day-name slicer, and personalized student details, the report effectively demonstrates advanced data visualization and analysis techniques.